

PROJECT PLANNING, DESIGN & DOCUMENTATION

REQUIREMENT ANALYSIS DOCUMENT

Project Title

LabStock AI – Intelligent Laboratory Inventory Management System

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1. PROJECT REQUIREMENT ANALYSIS OVERVIEW

1.1 Purpose of Requirement Analysis

Requirement Analysis is the process of identifying, understanding, documenting, and validating the requirements of a software system before development begins. It ensures that the development team clearly understands **what needs to be built, who will use it, what functions are required, and what conditions the system must satisfy.**

For **LabStock AI**, the requirement analysis focuses on developing a centralized system for managing laboratory inventory, monitoring stock levels, managing suppliers, and handling purchase orders.

1.2 Position in the Project Lifecycle

Requirement Analysis acts as a bridge between identifying the business problem and developing the final application.

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6
Business Problem	Requirement Analysis	System Design	Database Design	Agile Sprint Planning	Development
Identify the laboratory inventory problem and business needs.	Identify users, requirements, rules, and acceptance criteria.	Define system structure, modules, interfaces, and workflows.	Define data entities, attributes, and relationships.	Divide requirements into manageable and prioritized tasks.	Implement, integrate, test, and deliver the system.

Key Principle: A successful software project begins with understanding the problem and requirements before deciding on the technical solution.

2. UNDERSTANDING THE SYSTEM REQUIREMENTS

2.1 What is Requirement Analysis?

Requirement Analysis involves collecting and organizing the expectations of users and stakeholders and converting them into clear software requirements.

For LabStock AI, this process determines:

- What information needs to be stored and managed.
- What operations users should be able to perform.
- Which users require access to specific functions.
- How laboratory stock should be monitored.
- How suppliers and purchase orders should be handled.
- What security, performance, reliability, and usability standards the system should meet.

2.2 Why Requirement Analysis is Important

A laboratory may have a large number of supplies that need continuous monitoring. If inventory information is not properly organized, users may have difficulty knowing what is available, what needs to be reordered, or which supplier is associated with a particular item.

Requirement Analysis helps address these issues by converting general business needs into **specific, measurable, and testable requirements**. It also reduces confusion during development because each feature has a clearly defined purpose and expected behavior.

3. SYSTEM CASE STUDY

3.1 Business Problem

Laboratory inventory includes different supplies that must be available whenever required. Managing these supplies manually can make it difficult to maintain accurate stock records and track procurement activities.

The system is intended to address the following problems:

Inventory Management

The laboratory needs a centralized method for maintaining information about available supplies, their quantities, and other relevant inventory details.

Stock Monitoring

The system needs to identify supplies that reach their defined reorder level so that users can take appropriate action before the stock becomes unavailable.

Supplier Management

Supplier information needs to be maintained systematically so that users can easily identify and manage suppliers associated with laboratory purchases.

Purchase-Order Management

The laboratory needs to create and monitor purchase orders and track their status, including orders that have been only partially fulfilled.

3.2 Business Objective

The primary objective of **LabStock AI** is to develop a centralized laboratory inventory management system that improves the organization and monitoring of laboratory supplies.

The system aims to:

1. Maintain accurate laboratory inventory records.
2. Provide better visibility of current stock levels.
3. Identify supplies requiring replenishment.
4. Maintain organized supplier information.
5. Simplify purchase-order management and tracking.
6. Provide controlled access to system functionality.
7. Improve the overall efficiency of laboratory inventory management.

4. STAKEHOLDER & USER ANALYSIS

4.1 Stakeholders

A stakeholder is an individual or group that interacts with the system, benefits from it, manages it, or is affected by its operation.

Stakeholder	Role / Requirement
Laboratory Administrator	Responsible for maintaining inventory records, managing suppliers and purchase orders, and controlling user access.
Laboratory Staff	Requires access to inventory information to check the availability of laboratory supplies needed for daily activities.
Procurement Staff	Requires supplier and purchase-order information to manage the purchasing and replenishment process.
Laboratory Management	Requires visibility into inventory and procurement activities to ensure that essential supplies remain available.

5. ACTORS AND SYSTEM ROLES

An **actor** is a person or external entity that directly interacts with the software system.

Primary Actors

Administrator

The Administrator has access to management functions such as adding and updating supplies, managing suppliers, creating purchase orders, monitoring inventory, and managing users.

Laboratory User

The Laboratory User can access inventory information and perform functions permitted by their assigned role.

Actor–System Distinction

Technology components such as the database and application framework are part of the system infrastructure. They are not treated as human actors because they do not independently interact with the system as users.

6. FUNCTIONAL REQUIREMENTS

Functional requirements define **what the LabStock AI system must do**.

6.1 Inventory Management

- The system should allow authorized users to add new laboratory supplies with the required inventory information.
- Users with appropriate permissions should be able to view existing supply records and their current quantities.
- Authorized users should be able to edit or remove inventory records when required.
- The system should maintain updated stock information whenever inventory records are changed.

6.2 Stock Monitoring

- The system should allow a reorder point to be defined for laboratory supplies.
- The system should compare the available quantity with the defined reorder point.
- Supplies reaching or falling below the reorder point should be identified as low-stock items.
- The system should provide a clear indication that a low-stock item requires replenishment.

6.3 Supplier Management

- Authorized users should be able to add and maintain supplier information.
- Existing supplier information should be available for viewing and updating.
- Supplier records should be organized so they can be associated with procurement activities.

6.4 Purchase-Order Management

- Authorized users should be able to create purchase orders for required supplies.
- The system should store relevant purchase-order information.
- Users should be able to view and monitor the status of purchase orders.
- The system should support tracking of partially fulfilled orders.

6.5 User Management

- Administrators should be able to manage system users.

- Users should have appropriate roles and permissions.
- Restricted functions should only be accessible to authorized users.

6.6 Inventory Information

- The system should allow relevant product or supply images to be stored.
- Users should be able to view available inventory information according to their permissions.

7. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define **how well the system should operate** rather than specific functions.

Category	Requirement
Security	The system should protect restricted functions through authentication and role-based authorization.
Performance	Inventory, supplier, and purchase-order operations should be processed efficiently without unnecessary delays.
Usability	The interface should be simple, consistent, and easy for laboratory staff to understand and navigate.
Reliability	Inventory and procurement information should be stored accurately and remain consistent during system operations.
Scalability	The system should be capable of handling additional supplies, users, suppliers, and purchase orders as the laboratory grows.
Maintainability	The application should have a structured design that makes it easier to identify, modify, and extend functionality.
Availability	The system should remain accessible to authorized users whenever inventory and procurement activities need to be performed.

8. FUNCTIONAL VS NON-FUNCTIONAL REQUIREMENTS

Functional Requirement – WHAT?

Add a new laboratory supply.

Update inventory quantities.

Manage supplier information.

Create a purchase order.

Track order status.

Identify low-stock supplies.

Non-Functional Requirement – HOW?

Supply information should be stored securely.

Inventory operations should be performed efficiently.

The interface should be easy to use.

Purchase-order data should remain reliable.

The system should support future expansion.

The application should be maintainable.

Classification Examples

“The administrator should be able to add a laboratory supply.”

→ **Functional Requirement**

“The system should identify supplies below their reorder point.”

→ **Functional Requirement**

“Only authorized administrators should be able to manage users.”

→ **Non-Functional Requirement – Security**

“The system should process inventory operations efficiently.”

→ **Non-Functional Requirement – Performance**

9. USER STORIES

User stories express requirements from the perspective of the user. They follow the format:

As a [user], I want [functionality], so that [benefit].

Inventory Management

As an administrator, I want to add and manage laboratory supplies so that inventory records remain accurate and up to date.

Stock Monitoring

As an administrator, I want to monitor reorder levels so that I can identify supplies that need replenishment before they run out.

Supplier Management

As an administrator, I want to manage supplier information so that I can easily access supplier details during procurement.

Purchase Orders

As an administrator, I want to create and track purchase orders so that laboratory procurement activities can be monitored effectively.

Inventory Viewing

As a laboratory user, I want to view available supplies so that I can check whether the required laboratory materials are currently available.

10. ACCEPTANCE CRITERIA

Acceptance criteria define the conditions that must be satisfied for a requirement or user story to be considered complete.

10.1 Manage Laboratory Supply

No.	Acceptance Criterion
1	The user must be logged in and authorized to manage laboratory supplies before accessing the functionality.
2	Required supply information, including the supply name and quantity, must be entered correctly.
3	A reorder point should be defined so that the supply can be monitored effectively.
4	The system should validate the entered information before saving the record.
5	After successful submission, the supply should be stored and displayed in the inventory list.

10.2 Low-Stock Detection

No.	Acceptance Criterion
1	The supply must have a current quantity and defined reorder point.
2	The system should compare the available quantity with the reorder point.
3	A supply reaching or falling below the reorder point should be identified as low stock.
4	The system should clearly indicate that the supply requires replenishment.

11. GIVEN – WHEN – THEN SCENARIOS

Given–When–Then provides a simple way of describing how a requirement should behave.

Scenario 1: Adding a Laboratory Supply

Given an authorized administrator is logged into the system,

When the administrator enters valid laboratory supply information and submits the form,

Then the system should validate and save the information,

And the newly created supply should appear in the inventory list.

Scenario 2: Identifying Low Stock

Given a laboratory supply has a defined reorder point,

When its available quantity reaches or falls below that reorder point,

Then the system should identify the supply as low stock,

And indicate that replenishment is required.

12. REQUIREMENT-TO-DEVELOPMENT TRACEABILITY

Requirement analysis should provide a clear connection between the business requirement and the development work.

Project Stage	LabStock AI Example
Business Requirement	The laboratory requires a centralized system for managing supplies and procurement activities.
User Story	As an administrator, I want to manage laboratory supplies so that inventory remains accurate.
Acceptance Criteria	Supply information must be valid, successfully saved, and displayed in the inventory.
System Design	Design inventory modules, interfaces, workflows, and interactions between system components.
Database Design	Define entities such as supplies, suppliers, purchase orders, and users along with their relationships.
Development Task	Implement inventory management, stock monitoring, supplier management, and purchase-order functionality.
Sprint	Break the functionality into manageable development tasks and assign them to appropriate sprints.

13. REQUIREMENT ANALYSIS CHECKLIST

Before moving to system design, the following requirements should be confirmed:

✓ Business Requirements

The laboratory inventory problem, business objective, and expected benefits of the system have been clearly identified.

✓ User Requirements

The system users, their roles, responsibilities, and access permissions have been defined.

✓ Functional Requirements

The required inventory, stock monitoring, supplier, reorder, user management, and purchase-order functions have been documented.

✓ Non-Functional Requirements

Security, performance, usability, reliability, availability, scalability, and maintainability expectations have been established.

✓ Validation Requirements

The requirements are clear, measurable, testable, and supported by acceptance criteria and Given–When–Then scenarios.

14. EXPECTED OUTCOME

After implementing the identified requirements, **LabStock AI** is expected to provide a structured platform for laboratory inventory management.

The system should enable authorized users to:

- Maintain organized laboratory supply records.
- Monitor available stock and identify low-stock supplies.
- Manage supplier information efficiently.
- Create and track purchase orders.
- Monitor order fulfillment.
- Access inventory information according to user permissions.
- Maintain reliable and centralized inventory data.

15. NEXT PHASE OF THE PROJECT

The completed Requirement Analysis will serve as the input for the next project activities.

Current Phase	Next Phase	Following Phases
Requirement Analysis	System Design	Database Design → Agile Sprint Planning → Development → Testing

The next phase will focus on **how the identified requirements will be translated into the system architecture, modules, interfaces, workflows, and technical design of LabStock AI.**

